Onboarding Funnel Analysis

Identifying drop-offs and bottlenecks in user onboarding flow

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Objective & Approach

Dataset:

- onboarding_events with 3 fields: user_id, event_type, event_at
- 7 process steps from started_onboarding → opted_for_instant_transfer

Objectives:

- Identify where **drop-offs** occur in the funnel
- Highlight **pain points** by analyzing time to move between steps

Methodology:

- Performed data quality checks (nulls, duplicates, event typos)
- Built event sequence per user
- Calculated drop-off counts, drop-off % and conversion % per step
- Computed step-to-step times (Median, Avg, P90) to detect bottlenecks

Code Review & Query Logic Overview

Data Quality Checks – Verified for nulls, duplicates, and event-type inconsistencies.

Data Cleaning – Converted *event_at* from text to timestamp for time calculations and corrected event name typo ("started_onboading" → "started_onboarding").

- **Drop-Off Query**
  - events_clean CTE – Standardized event names and timestamps.
  - drop_off_count_cte – Counted distinct users per event; assigned step_order using CASE WHEN.
  - prev_count_cte – Used LAG() to retrieve previous step's user count.
  - **Final Query** – Selected step_order, event_type, and count_of_users; calculated:
    - *Drop-off count* = previous − current
    - *Drop-off rate* = (previous − current) / previous × 100
    - *Conversion rate* = current / previous × 100 (for validation)
- Output produced step-wise user counts, drop-off %, and conversion %

- **Time- Taken Query**
  - events_clean CTE – Same as drop off
  - step_order CTE – Assigned step sequence using CASE WHEN
  - time_difference CTE – Used LEAD(event_time) and LEAD(step_order) to get next step's timestamp and order.
  - deltas CTE – Computed seconds between consecutive steps; filtered negatives to ensure valid progression.
  - **Final Query** – Aggregated by step_pair; calculated:p50 (median), p90, average, min, and max time taken between steps.
  - Output summarized time distributions (in seconds/minutes/hours) for each transition.

- Each step analyzed using distinct users to avoid double-counting repeated events.
- Median (p50) and p90 time chosen for robustness against skew and to surface tail delays.
- Sequential event order validated by timestamps; only consecutive transitions analyzed.

Code Output

Drop Off output

step_order	event_type	count_of_users	drop_count	drop_off_rate	conversion_rate
1	started_onboarding	10000	NULL	NULL	NULL
2	accepted_terms_of_serv	9398	602	6.02	93.98
3	passed_kyc	9234	164	1.75	98.25
4	started_application	8206	1028	11.13	88.87
5	submitted_application	7889	317	3.86	96.14
6	confirmed_advance	7514	375	4.75	95.25
7	opted_for_instant_trans	7352	162	2.16	97.84

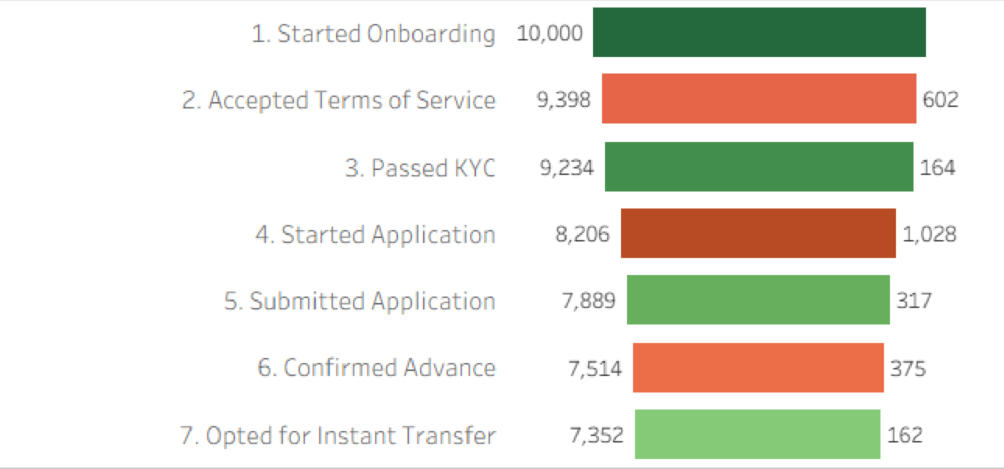
Time Taken output

step_pair	p50_seconds	p90_minutes	avg_hours	fastest_seconds	slowest_hours	n_users
started_onboarding - accepted_terms_of_service	13	1.03	6.75	6	760.45	9398
accepted_terms_of_service - passed_kyc	0	0	0.7	0	724.57	6108
passed_kyc - started_application	17.19	4338.45	30.64	4.35	988.35	8165
started_application - submitted_application	18.46	8.65	8.02	0.29	845.61	7858
submitted_application - confirmed_advance	24	3.92	9.38	3	867.9	7514
confirmed_advance - opted_for_instant_transfer	10	0.38	5.7	4	982.93	7352

Drop-off Analysis → Where are users churning?

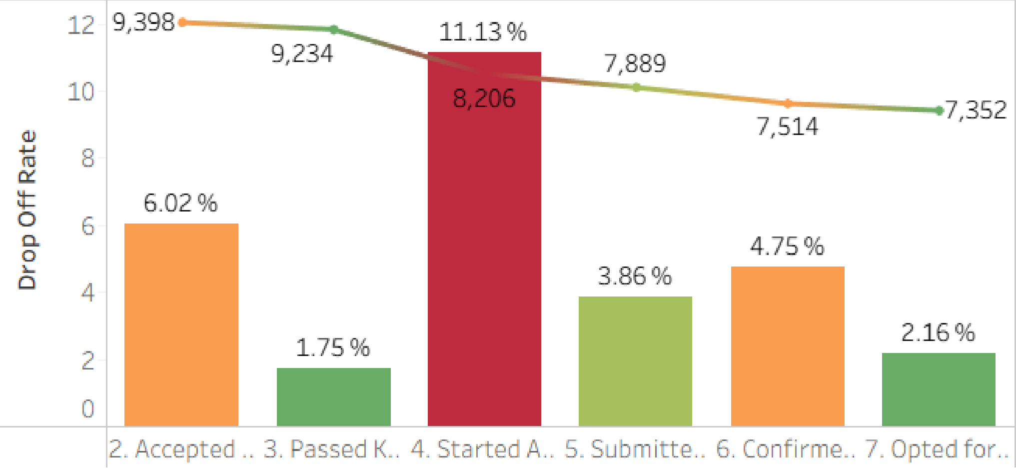
Event Type - User count & Drop-off count Funnel

Biggest Drop off - 1028 users at 'Started Application'



Drop off Rate by Event Type

Biggest drop-off: 11% at 'Started Application 'stage



Key Takeaways:

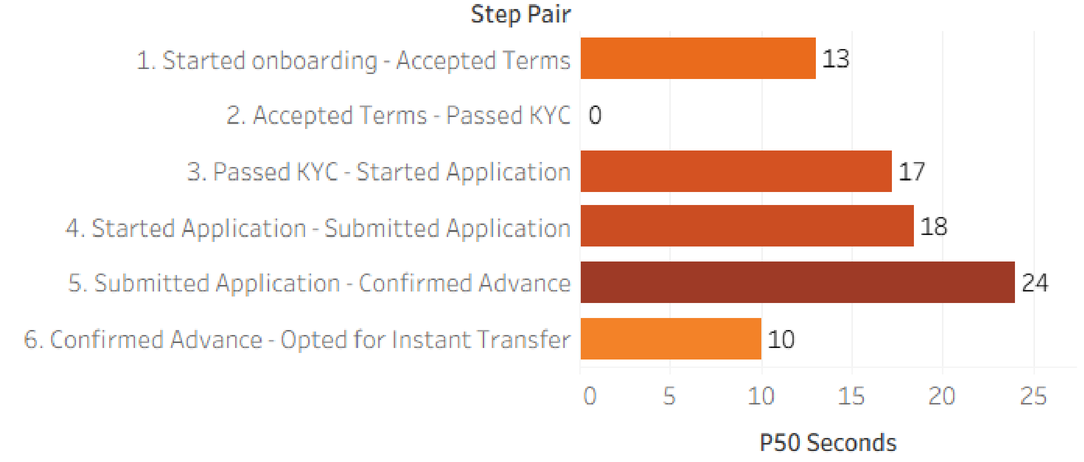
- **Biggest drop-off:** Started Application (11%) → users are abandoning after passing KYC to starting application stage
- **Biggest delay:** Passed KYC → Started Application (median 17s, avg 31 hrs, P90 ~72 hrs).
- **Recommendation:** Simplify KYC-to-application handoff and introduce reminders/push nudges to reduce abandonment

Time Taken through Each Step → How long does it take users to move forward?

*Units chosen for readability: seconds (median, fastest), minutes (p90), hours (avg/slowest)

How long it takes 'typical' (median or 50%) users to move through each step?

Bottleneck - Submitted Application to Confirmed advance - 24 seconds



P50 v/s P90 Comparison - Time taken for 50% v/s 90% users to move through steps

Biggest bottleneck by 90% users: Passed KYC - Started Application (4,338 minutes ~ 72 hours)

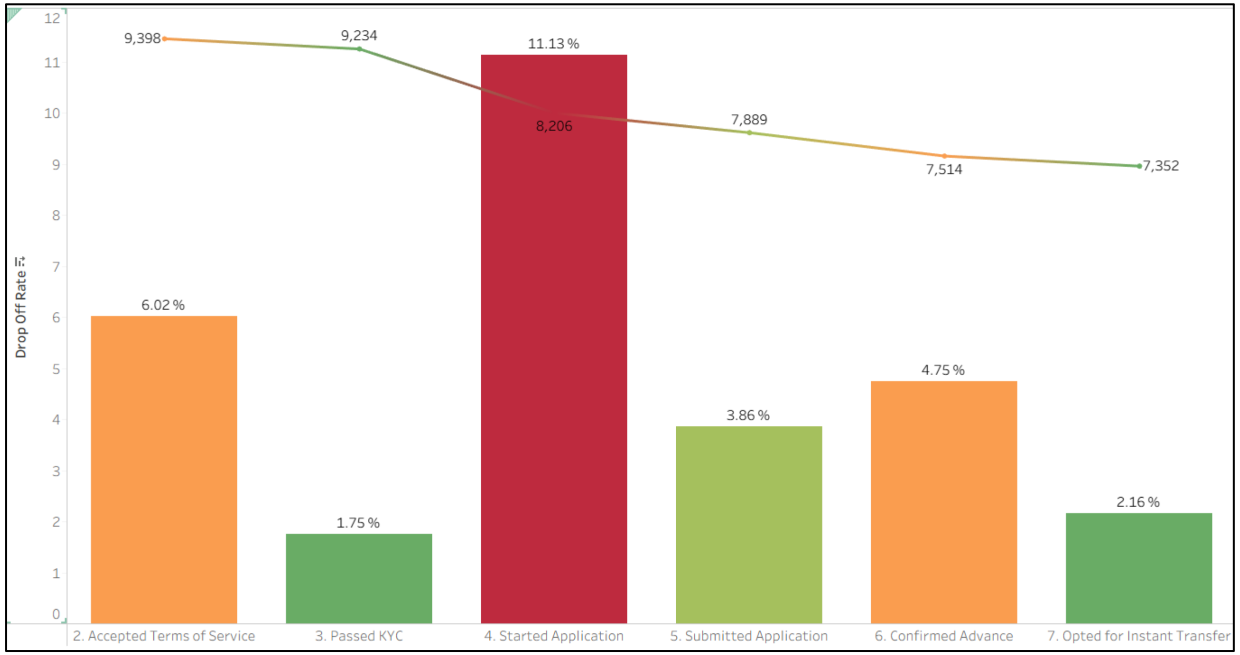
Step Pair	P50 Seconds	P90 Minutes
1. Started onboarding - Accepted Terms	13	1
2. Accepted Terms - Passed KYC	0	0
3. Passed KYC - Started Application	17	4,338
4. Started Application - Submitted Application	18	9
5. Submitted Application - Confirmed Advance	24	4
6. Confirmed Advance - Opted for Instant Transfer	10	0

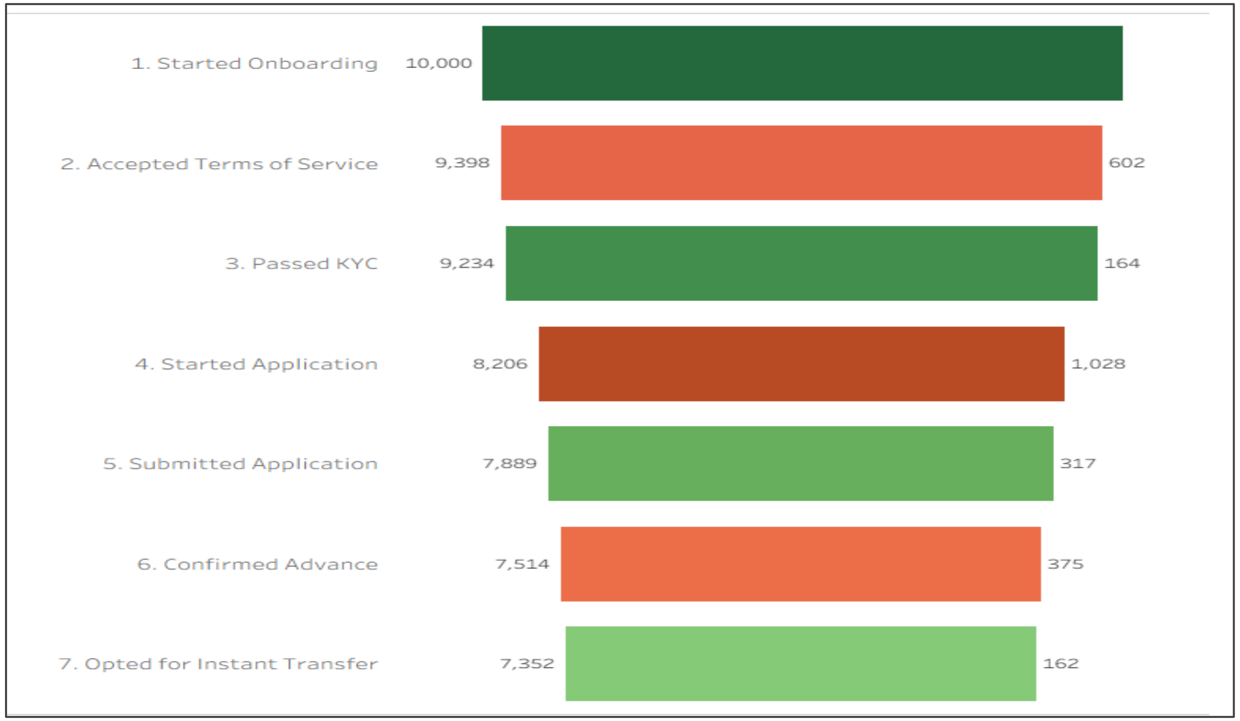
Fastest completion v/s Slowest completion

	Fastest Seconds	Slowest Hours	Avg Hours
1. Started onboarding - Accepted Terms	6.0	760.5	6.8
2. Accepted Terms - Passed KYC	0.0	724.6	0.7
3. Passed KYC - Started Application	4.4	988.4	30.6
4. Started Application - Submitted Application	0.3	845.6	8.0
5. Submitted Application - Confirmed Advance	3.0	867.9	9.4
6. Confirmed Advance - Opted for Instant Transfer	4.0	982.9	5.7

Drop Off Analysis - Where Are Users Churning?

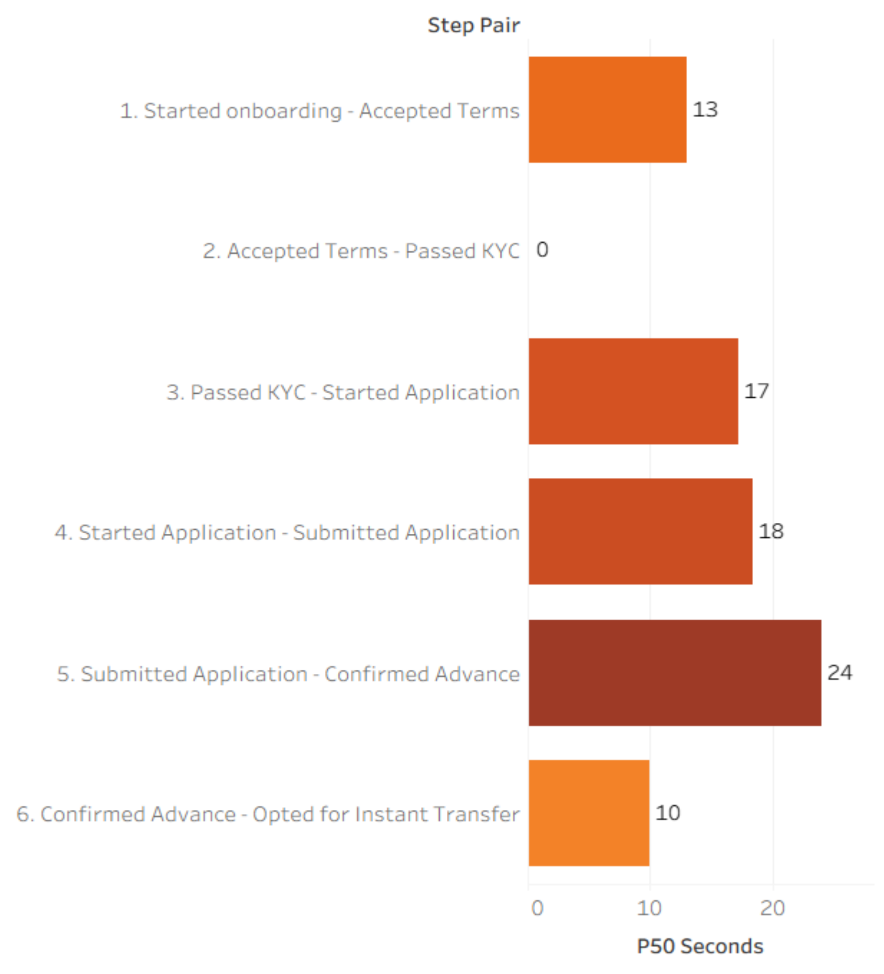
- Biggest drop-off: Passed KYC → Started Application – 11% (1,028 users)
- Second largest drop-off: Started Onboarding → Accepted Terms of Service – 6% (602 users)
- Smallest drop-off: Accepted Terms → Passed KYC – 1.75% (164 users)
- Pattern suggests friction after KYC completion but before starting the application
- Overall funnel conversion: ~74% from start to finish





Time Analysis – Pain Points

- Median (p50): Most users complete steps in seconds
  - Fastest: Accepted Terms → Passed KYC (~0s)
  - Slowest: Submitted Application → Confirmed Advance (~24s)
- p90 outlier: **Passed KYC → Started Application** takes up to 4,338 min (~72h)
  - This step also has the highest drop-off → clear bottleneck
- Other step with notable gap: **Started Application → Submitted Application**
  - Median: 18s vs P90: 9 min (much smaller than KYC → Application)
- Averages are skewed by outliers
  - Confirmed by fastest vs slowest comparisons (seconds → hundreds of hours)



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Recommendations

- **Simplify the KYC → Application handoff** by making the transition clearer and more seamless, ensuring that users who complete KYC are automatically guided into starting their application.
- Optimize the application interface to minimize abandonment at the "Started Application" step. Complement UI improvements with targeted user feedback surveys to identify friction points.
- Send proactive reminders (push notifications or emails) to re-engage users inactive for more than 24 hours, reducing drop-offs due to inactivity.
- Continuously monitor funnel health by tracking drop-off %, end-to-end conversion, and time-to-completion (p50/p90). Regular feedback loops will ensure changes align with user needs.
- **Final Recommendation:** Prioritize fixing the KYC → Application bottleneck first — it represents both the largest source of drop-off and the longest delay. After improvement, track impact across funnel metrics and iterate on other pain points as needed.